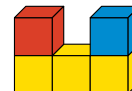


CANADIAN MATH KANGAROO CONTEST PROBLEMS

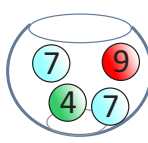
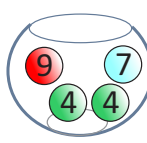
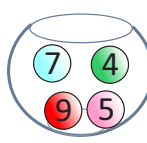
PART A: EACH CORRECT ANSWER IS WORTH 3 POINTS

1. The figure shows 5 cubes viewed from the front.
What is the view from above?



- (A)  (B)  (C)  (D)  (E) 

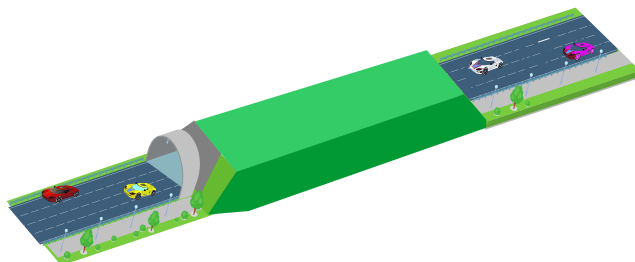
2. Each bowl contains four numbered balls, as shown. In which bowl is the sum of all the numbers largest?

- (A)  (B)  (C)  (D)  (E) 

3. This is my grandfather's birthday cake.
A large candle stands for 10 years and a small one for 1 year.
How old is my grandfather?

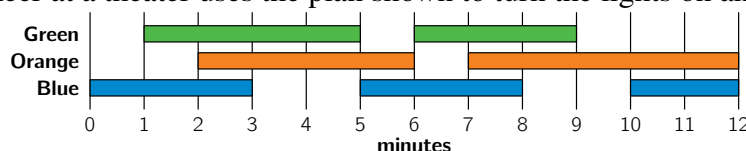


- (A) 65 (B) 66 (C) 76 (D) 77 (E) 78
4. Pablo puts 10 toy cars on this racetrack.



How many cars are in the tunnel?

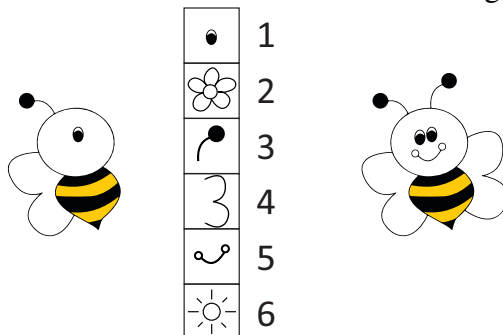
- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9
5. A lighting engineer at a theater uses the plan shown to turn the lights on and off.



How long in total are exactly 2 of the lights on at the same time?

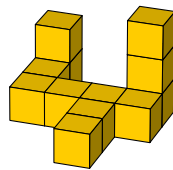
- (A) 2 minutes (B) 6 minutes (C) 8 minutes (D) 9 minutes (E) 10 minutes

6. Raha wants to complete the illustration of the bee on the left according to the model on the right.



Raha needs to win points to unlock parts of the bee.

How many points does she need to win to complete the bee?

- (A) 9 (B) 10 (C) 11 (D) 12 (E) 13
7. A student made the shape shown using 12 cubes.
He put one drop of glue between any two cubes that share a common face.
How many drops of glue did he use?
- 
- (A) 8 (B) 9 (C) 10 (D) 11 (E) 12
8. A wooden fence consists of a series of vertical planks, each joined to the next post by four horizontal planks. The first and last planks in the fence are vertical. Which of the following numbers could be the total number of planks in the fence?
- (A) 95 (B) 96 (C) 97 (D) 98 (E) 99
9. The letters a and b are to be replaced by positive integers so that the equation $\frac{a}{5} = \frac{7}{b}$ is correct.
In how many different ways can this be done?
- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4
10. What is the units digit of the product $(5^5 + 1)(5^{10} + 1)(5^{15} + 1)$?
- (A) 1 (B) 2 (C) 4 (D) 5 (E) 6

PART B: EACH CORRECT ANSWER IS WORTH 4 POINTS

11. The table has 30 boxes.

Lauren painted the boxes in row 3, row 6, column C and column D.

How many boxes are not painted?

(A) 8

(B) 10

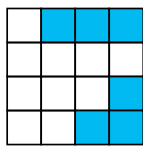
(C) 12

(D) 18

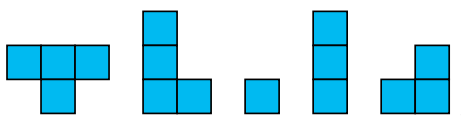
(E) 22

	A	B	C	D	E
1					
2					
3					
4					
5					
6					

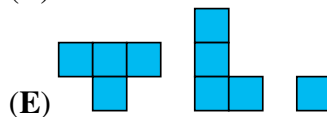
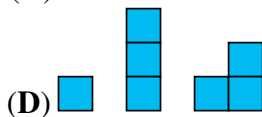
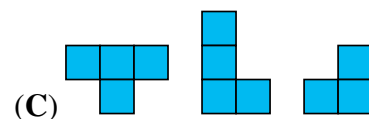
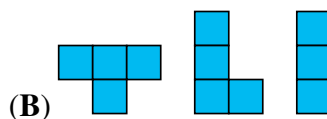
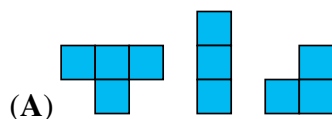
12. Max wants to complete this puzzle



He has these 5 different pieces



Which pieces does he have to use to complete the puzzle?



13. Adam and Brenda have 9 marbles each. Together, they have 8 red and 10 blue marbles. Brenda has twice as many blue marbles as red marbles. How many blue marbles does Adam have?

(A) 3

(B) 4

(C) 5

(D) 6

(E) 0

14. There are 8 cars waiting in a queue for the ferry. Every car contains either 2 or 3 people. There are 19 people in total waiting for the ferry. How many cars contain exactly 2 people?

(A) 2

(B) 3

(C) 4

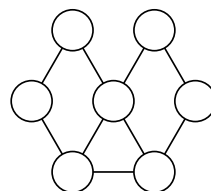
(D) 5

(E) 6

15. Hatice wants to paint the circles in the figure.

She wants to paint any two circles connected with a line of different colours.

What is the smallest number of colours she needs?



(A) 2

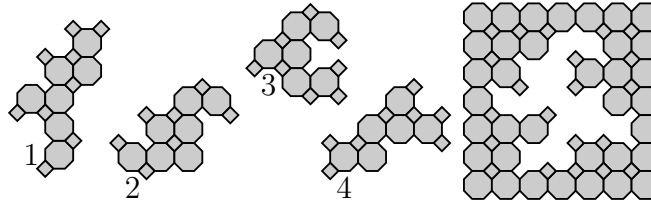
(B) 3

(C) 4

(D) 5

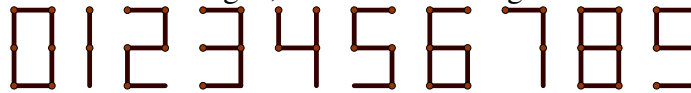
(E) 6

16. Which two tiles should be used to complete the puzzle?



- (A) 1 and 2 (B) 1 and 4 (C) 2 and 3 (D) 2 and 4 (E) 3 and 4

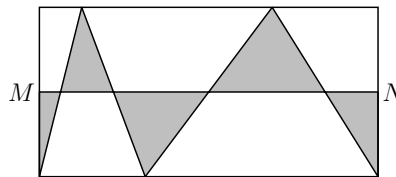
17. Matchsticks can be used to write digits, as shown in the figure.



How many different positive integers can be written using exactly six matchsticks in this way?

- (A) 2 (B) 4 (C) 6 (D) 8 (E) 9

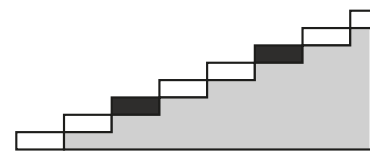
18. Points M and N are the midpoints of two sides of the rectangle.



What fraction of the area of the rectangle is shaded?

- (A) $\frac{1}{6}$ (B) $\frac{1}{5}$ (C) $\frac{1}{4}$ (D) $\frac{1}{3}$ (E) $\frac{1}{2}$

19. Every third step of a staircase with 2023 steps is coloured black. The first seven steps are shown in the figure. Anita walks up the steps one at a time, starting with either her right or left foot, alternating each step.



What is the smallest number of black steps she will step on with her right foot?

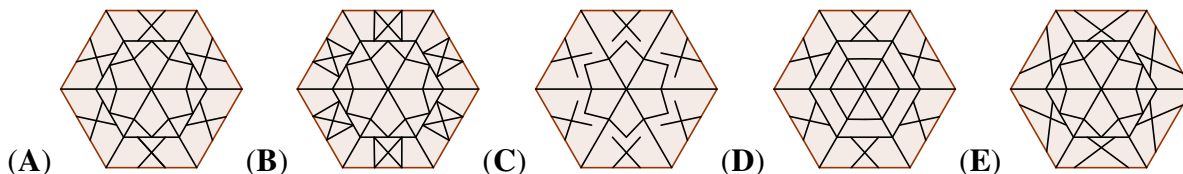
- (A) 0 (B) 333 (C) 336 (D) 337 (E) 674

20. We are given five numbers a_1, a_2, a_3, a_4, a_5 whose sum is S . For each $k, 1 \leq k \leq 5$, we know that $a_k = k + S$. What is the value of S ?

- (A) $\frac{15}{4}$ (B) $-\frac{15}{4}$ (C) -15 (D) 15
(E) none of the previous

PART C: EACH CORRECT ANSWER IS WORTH 5 POINTS

21. Elvis has 6 identical triangles like this . Which of the following figures can he make?



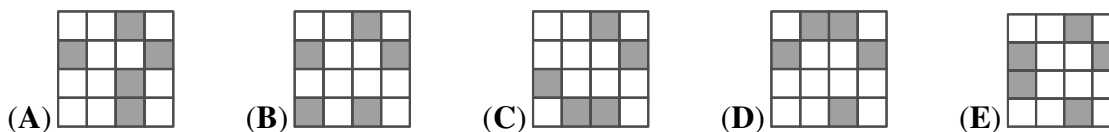
22. Five children share a birthday and each child has their own cake. Lea is two years older than Jose, but one year younger than Ali. Vittorio is the youngest. Which is Sarah's cake?



23. Three frogs live in a pond. Each night, one of the frogs sings a song to the other two. After 9 nights, one of the frogs had sung 2 times. Another frog had listened to 5 songs. How many songs had the third frog listened to?

(A) 7 (B) 6 (C) 5 (D) 4 (E) 3

24. Maria has shaded exactly 5 cells in a 4×4 grid. She challenges 5 of her friends to guess which cells she has shaded. The grids they have drawn are shown below. Maria looks at them and says: "One of you is right and each of the rest of you has four cells correct." Which is the correct answer?



25. Martin has three cards with numbers written on both sides. The card with the number 1 on one side has the number 4 on the other side, the card with the number 2 has 5 on the opposite side, and the card with the number 3 has 6 on the opposite side. Martin randomly places three cards on the table and adds up the three numbers he sees. How many different sums can Martin get?

	Front	Back
Card 1	1	4
Card 2	2	5
Card 3	3	6

(A) 3 (B) 4 (C) 5 (D) 6 (E) 10



26. Bart wrote the number 1015 as a sum of numbers using only the digit 7. He used the digit 7, a total of 10 times, as shown. Now he wants to write the number 2023 as a sum of numbers using only the digit 7, a total of 19 times. How many times will he use the number 77?

777
77
+ 77
77
7
1015

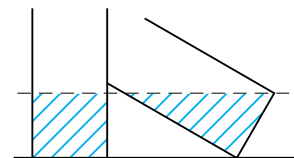
- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6
27. The number 5^{5^6} can be written in the form n^n for some integer n . What is the value of n ?
- (A) 5^{30} (B) 5^6 (C) 5^5 (D) 30 (E) 11

28. In how many different ways can the word **BANANA** be read from the table by moving from one cell to another cell with which it shares an edge? (Cells may be visited more than once.)

B	A	N
A	N	A
N	A	N

- (A) 14 (B) 28 (C) 56 (D) 84 (E) other value
29. Two functions f and g on $\mathbb{R}$ satisfy the system of equations
$$f(x) + 2g(1 - x) = x^2 \quad \text{and} \quad f(1 - x) - g(x) = x^2.$$
What is f ?
- (A) $x^2 - \frac{4}{3}x + \frac{2}{3}$ (B) $x^2 + \frac{4}{3}x + \frac{2}{3}$ (C) $-x^2 - \frac{4}{3}x + \frac{2}{3}$
(D) $x^2 - 4x + 5$ (E) there is no such function

30. Two identical cylindrical water tanks contain the same amount of water. One cylinder is standing upright, and the other is leaning against it, and the water level in each of them is the same as in the picture. The bottom of each of the cylinders is a circle with area $3\pi m^2$. How much water does each tank contain?



- (A) $3\sqrt{3}\pi m^3$ (B) $6\pi m^3$ (C) $9\pi m^3$ (D) $\frac{3\pi}{4} m^3$
(E) it's impossible to determine from the information given

Answer keys: BACBCEDBEE CABDBDCCDB ACBEBECDAC